Graduation Project Report

ALTINBAŞ UNIVERSITY Faculty of Engineering and Architecture Software Engineering Department

Turkish Customer Review Classification Service

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GitHub Repository: https://github.com/silasea/turkish-review-classifier

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1. Introduction

E-commerce platforms generate substantial amounts of textual customer feedback through product reviews. Analyzing this information can help businesses understand customer satisfaction, evaluate product quality, and identify recurring issues. Natural Language Processing (NLP) provides techniques for automatically processing and classifying such text. This project focuses on developing a Turkish customer review classification service that combines machine learning, an API, automated testing, and containerization.

2. Problem Statement

Manually analyzing large volumes of customer reviews requires considerable time and effort. Manual classification can also be inconsistent, making it difficult to identify patterns across thousands of reviews. Consequently, businesses may overlook important complaints or fail to recognize changes in customer sentiment. An automated system is needed to classify Turkish customer reviews consistently and efficiently.

3. Motivation

Customer feedback is an important source of information for improving products and services. Efficient review analysis can help businesses recognize customer concerns and make more informed decisions. From a software engineering perspective, this project also provides an opportunity to develop an end-to-end NLP application that combines model training, API development, testing, and reproducible deployment.

4. Project Objective

The primary objective is to develop a working NLP service that classifies Turkish e-commerce reviews by sentiment. The project will establish a TF-IDF and Logistic Regression baseline and compare it with a fine-tuned Turkish language encoder. The models will be evaluated using appropriate classification metrics. The selected model will then be integrated into a FastAPI service with a simple user interface, automated tests, and Docker-based packaging.

5. Preliminary Scope

The project will include data preprocessing, baseline model development using TF-IDF and Logistic Regression, fine-tuning a Turkish language encoder, model evaluation, REST API development with FastAPI, a simple user interface, automated testing with pytest, and Docker-based containerization. SQLite will be used for storing sentiment analysis results.

The initial version will focus exclusively on Turkish customer reviews and sentiment classification. User authentication, account management, multilingual support, and cloud deployment are outside the initial scope and may be considered in future work.

6. Dataset

The project will use the MüşteriYorumları dataset, which contains approximately 103,000 Turkish e-commerce reviews from Hepsiburada and Trendyol with ratings from one to five stars. The dataset will be examined to assess its structure, label distribution, and data quality. Star ratings will be mapped to three sentiment classes: ratings of 1-2 as Negative, 3 as Neutral, and 4-5 as Positive. This mapping will be verified during dataset inspection by examining representative reviews from each rating level.

7. Requirements Analysis

The primary user of this project is a customer experience team member working for an e-commerce business. This user needs to analyze customer reviews to understand customer sentiment and identify negative feedback. Manually reading and classifying large numbers of reviews can be time-consuming and make it difficult to identify negative comments efficiently.

The proposed system will provide a Turkish customer review sentiment classification service. Its main objective is to predict the sentiment class of a Turkish review and return the result to the user. The first prototype will focus on three essential capabilities: accepting review text, classifying the text using a machine learning model, and displaying the predicted sentiment. A simple and functional interface may be included to demonstrate this workflow. More advanced interface design, bulk review uploads, and analysis history will be postponed until the core classification process works reliably. The dataset will be finalized after its suitability and data quality have been evaluated.

8. Project Scope

In Scope

- Accepting Turkish customer reviews as text input.
- Classifying reviews into Positive, Neutral, or Negative sentiment classes.
- Displaying predictions through a FastAPI-based REST API and a simple user interface.
- Comparing a TF-IDF and Logistic Regression baseline with a fine-tuned Turkish language model.
- Evaluating model performance using classification metrics, including macro-F1.
- Storing analysis results in an SQLite database.
- Testing and packaging the application using pytest and Docker.

Out of Scope

- User authentication and account management.
- Multilingual sentiment classification.
- Bulk review uploads and advanced review analytics.
- Cloud deployment in the initial version.

9. Functional Requirements

ID	Requirement statement	Priority
FR-01	The system shall accept a Turkish customer review as text input.	Must
FR-02	The system shall classify an input review into one of three sentiment classes: Positive, Neutral, or Negative.	Must
FR-03	The system shall display the predicted sentiment class to the user.	Must
FR-04	The system shall reject empty or whitespace-only review input and display a clear validation message.	Must
FR-05	The system shall expose the sentiment classification functionality through a REST API implemented with FastAPI.	Must
FR-06	The system shall store each analysis result in an SQLite database for later retrieval.	Should

10. Non-Functional Requirements

ID	Quality	Requirement	How to verify
NFR-01	Performance	The system shall return a sentiment prediction within 2 seconds for a single review of up to 500 characters, excluding initial model loading, on the specified development laptop.	Measure response time across 30 individual requests and check whether each prediction meets the target.
NFR-02	Classification quality	The selected model shall achieve a macro-F1 score of at least 0.75 on a held-out test set containing all three sentiment classes.	Evaluate the trained model on a test set that was not used for training or model selection, and calculate macro-F1.
NFR-03	Input validation	The system shall return an appropriate validation error for empty or whitespace-only input without attempting sentiment classification.	Submit empty and whitespace-only inputs through the API and verify the error response and that no prediction is produced.

11. Constraints and Assumptions

Constraints

CON-01 — Development Time: The project must be developed within the nine-week graduation project schedule. Therefore, development will prioritize a working prototype and essential functionality over advanced features.

CON-02 — Computational Resources: Model training and fine-tuning will be limited by the available hardware and computational resources. Free cloud GPU services may be used when available, but their availability and usage limits cannot be guaranteed.

Assumptions

ASM-01 — Dataset Label Quality: The selected dataset is assumed to contain sufficiently reliable review ratings for deriving Positive, Neutral, and Negative sentiment labels. This assumption will be examined by inspecting the dataset, its rating distribution, and representative reviews before finalizing the label mapping.

12. Use Cases

US-01: Classify a Customer Review

As a customer experience team member, I want to submit a Turkish customer review and see its predicted sentiment class, so that I can identify the general sentiment expressed in the review.

Related Requirements: FR-01, FR-02, FR-03

Acceptance Criteria

1. Given a valid Turkish customer review, when the user submits the review, then the system shall classify it as Positive, Neutral, or Negative.
2. Given that the classification process has completed, when the system returns the prediction, then the predicted sentiment class shall be displayed to the user.

US-02: Handle Empty Review Input

As a customer experience team member, I want to receive a clear validation message when I submit an empty review, so that I can correct my input before requesting a sentiment prediction.

Related Requirements: FR-04, NFR-03

Acceptance Criteria

1. Given an empty or whitespace-only review, when the user submits it, then the system shall reject the input and display a clear validation message.
2. Given an empty or whitespace-only review, when the API receives the request, then it shall return an appropriate validation error without generating a sentiment prediction.

13. Development Plan

Week	Planned Deliverable
Week 3	Design the system architecture and document two Architecture Decision Records (ADRs): one explaining the choice of SQLite and the other explaining the selection of BERTurk and the use of a common model interface.
Week 4	Develop an end-to-end baseline using TF-IDF and Logistic Regression. Load and preprocess the dataset, train the baseline model, and create a basic FastAPI service. Integrate the baseline model into the API and demonstrate sentiment prediction through an API request.
Week 5	Fine-tune BERTurk for three-class Turkish sentiment classification. Integrate it into the existing FastAPI service by replacing the baseline model behind the same prediction interface. Verify that the API continues to work with the new model.

Week	Planned Deliverable
Week 6	Implement automated tests using pytest and prepare a comparison table for the baseline and BERTurk models using macro-F1, accuracy, precision, and recall. Evaluate both models on the same held-out test set.
Week 7	Containerize the application using Docker, maintain a requirements.txt file, and verify that the application runs successfully in a clean environment using documented setup instructions.
Week 8	Complete the graduation project report and improve the GitHub README with the project overview, features, scope, technology stack, installation instructions, usage examples, and evaluation results.
Week 9	Prepare and deliver the final project demonstration, showing the end-to-end application workflow, sentiment predictions, model evaluation results, and key engineering decisions.

14. Methodology

Modeling Approach

The project will follow a simple-to-complex modeling approach. First, a TF-IDF and Logistic Regression model will be developed as a baseline, followed by a fine-tuned BERTurk model. Both models will be accessed through a common interface in the Model Layer, allowing the model to be replaced without changing the API. Their performance will be evaluated using the same test set to ensure a fair comparison.

Justification

This approach is suitable for a one-person project with a nine-week timeline and limited GPU resources. The TF-IDF and Logistic Regression baseline can be implemented quickly and does not require a GPU, making it possible to build a working system early. The more complex BERTurk model can then be developed as the next step. Even if fine-tuning takes longer than expected, the project will still have a functional baseline system.

15. System Architecture

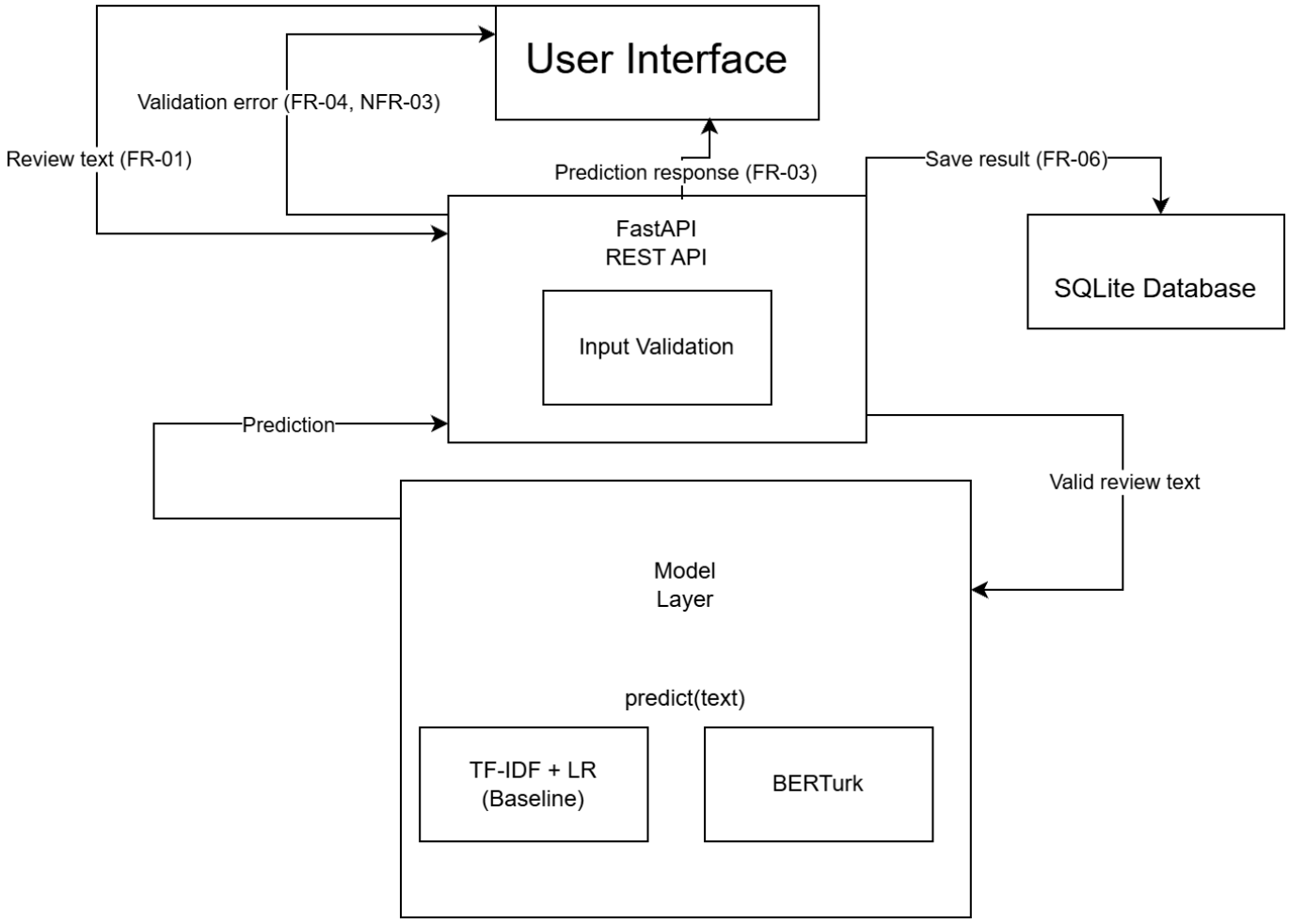


Figure 3.1: Proposed system architecture showing the main components and the flow of a review from input to prediction.

Figure 3.1 presents the proposed architecture. The User Interface sends the review text to the FastAPI REST API, where input validation checks whether the text is empty or contains only whitespace. Valid text is forwarded to the Model Layer, which produces a sentiment prediction through the common predict(text) interface. The prediction is returned to the API, which stores the result in the SQLite database and returns the predicted class to the User Interface. If validation fails, a validation error is returned to the user and the Model Layer is not called.

15.1 Component Responsibilities

Component	Responsibility / main input	Output / connection
User Interface	Allows the user to enter a Turkish customer review.	Sends the review text to the FastAPI REST API and displays the predicted sentiment or validation error returned by the API.
FastAPI REST API	Receives review text from the User Interface and manages the classification request. Includes input validation to reject empty or whitespace-only text.	Sends valid text to the Model Layer; receives the predicted sentiment; stores the analysis result in SQLite; returns the prediction or a validation error to the User Interface.

Component	Responsibility / main input	Output / connection
Model Layer	Processes valid Turkish review text using the selected sentiment classification model. Provides a common predict(text) interface for both TF-IDF + Logistic Regression and BERTurk.	Produces a predicted sentiment class (Positive, Neutral, or Negative) and sends it back to the FastAPI REST API.
SQLite Database	Stores analysis results received from the FastAPI REST API, including the review text, predicted sentiment class, and timestamp.	Stores the analysis records for later retrieval by the application.

System Boundaries

The entire system will run locally on a single machine during development and demonstration. The User Interface, FastAPI REST API, Model Layer, and SQLite Database will operate within this local environment. The main external dependency is the pre-trained language model downloaded from Hugging Face; its availability and hosting are outside the project's direct control. Once downloaded, the model will run locally during application execution.

15.2 End-to-End Execution Scenario

Step	Component	Input, operation, and output
1	User Interface	The user enters a Turkish customer review and submits it to the FastAPI REST API.
2	FastAPI REST API	Receives the review text through the /predict endpoint and forwards it to input validation.
3	Input Validation (within FastAPI)	Checks whether the review is empty or contains only whitespace. If valid, forwards the text to the Model Layer.
4	Model Layer	Processes the review using the selected classification model through the common predict(text) interface and returns a sentiment class to the API.
5	FastAPI REST API / SQLite Database	Receives the predicted sentiment and stores the review text, predicted class, and timestamp in SQLite.
6	FastAPI REST API / User Interface	Returns the prediction as an API response. The User Interface displays the predicted sentiment to the user.

Example API Endpoint

Endpoint: `POST /predict`

Request body:

```
{
  "text": "Ürün çok kaliteli ve hızlı teslim edildi."
```

```
}
```

Successful response (HTTP 200 OK):

```
{
  "sentiment": "Positive"
}
```

The `sentiment` field can contain one of three values: Positive, Neutral, or Negative.

Error Scenario: Empty Review Input

If the user submits an empty review or a review containing only whitespace, the FastAPI input validation mechanism rejects the request before it reaches the Model Layer.

Example request:

```
{
  "text": "    "
}
```

Error response (HTTP 422 Unprocessable Entity):

```
{
  "detail": "Review text cannot be empty."
}
```

In this scenario the Model Layer is not called, no sentiment prediction is generated, no analysis result is stored in SQLite, and the validation error is returned to the User Interface and displayed to the user.

16. Requirements-to-Design Mapping

ID	Requirement in my project	Design response / component
FR-02	Classify a Turkish customer review into three sentiment classes: Positive, Neutral, or Negative.	Model Layer, using a common predict(text) interface implemented by TF-IDF + Logistic Regression and BERTurk.
FR-04	Reject empty or whitespace-only review input and display a clear validation message.	FastAPI REST API, using input validation before forwarding the request to the Model Layer.
FR-06	Store each analysis result in an SQLite database for later retrieval.	SQLite Database, connected to the FastAPI REST API to store the review text, predicted sentiment class, and timestamp.

ID	Requirement in my project	Design response / component
NFR-01	Return a sentiment prediction within 2 seconds for a single review of up to 500 characters, excluding initial model loading.	Model Layer and FastAPI REST API. The selected model will be loaded into memory and kept available for subsequent requests instead of being reloaded for every prediction.
NFR-02	Achieve a macro-F1 score of at least 0.75 on a held-out test set containing all three sentiment classes.	Model Layer. Both classification models will be evaluated using the same held-out test set, and macro-F1 will guide model selection.

The macro-F1 value stated in NFR-02 is a project target rather than a measured result. The two-second limit in NFR-01 excludes the initial model loading time.

17. Technology Selection

ADR-01: Database Selection — SQLite vs PostgreSQL

Field	Decision
Issue	Which database technology should be used to store sentiment analysis results?
Selected option	SQLite
Alternative	PostgreSQL
Reason / trade-off	SQLite was selected because the project is a small-scale, locally running prototype with a limited number of users and a relatively small amount of analysis data. It does not require a separate database server, user or server configuration, or additional deployment infrastructure. This reduces setup complexity and allows the database to run directly within the application environment. PostgreSQL offers better support for concurrent users, larger workloads, and more complex database requirements. However, these capabilities are not essential for the current project scope. The main trade-off is that SQLite may become less suitable if the system needs to support many simultaneous users or significantly larger workloads in the future.
Reference	https://www.sqlite.org/whentouse.html

ADR-02: Language Model Selection — BERTurk vs mBERT

Field	Decision
Issue	Which pre-trained language model should be selected for Turkish customer review sentiment classification?
Selected option	BERTurk (Turkish-specific BERT model)
Alternative	mBERT (Multilingual BERT)

Field	Decision
Reason / trade-off	BERTurk was selected because it is specifically pre-trained for Turkish and is better suited to learning Turkish linguistic structures and contextual features. İncidelen and Aydoğan (2025) compared Turkish-specific and multilingual language models for sentiment analysis using the TRSAv1 dataset. Their findings indicate that Turkish-specific models, particularly BERTurk, generally perform better than multilingual alternatives. mBERT supports multiple languages and may be more suitable for multilingual applications, but multilingual coverage is not required in this project. The trade-off is that BERTurk is specialized for Turkish, whereas mBERT offers broader language coverage. The final model performance will still be evaluated on the project's own dataset.
Reference	https://doi.org/10.36222/ejt.1592448

Selecting BERTurk at this stage is an initial technical choice supported by published research, not a conclusion based on this project's own experimental results. The final model selection will be verified using macro-F1 results obtained on the project's own dataset.

18. Design Decisions and Risks

Risk	Potential impact	Mitigation / verification
RISK-01: Difficulty distinguishing the Neutral class	Three-star reviews may contain mixed or ambiguous opinions, making it difficult for the model to distinguish Neutral sentiment from Positive or Negative sentiment. This may reduce classification performance, especially macro-F1.	Inspect representative three-star reviews and evaluate class-wise precision, recall, and F1-score. Review the rating-to-sentiment mapping during preprocessing and examine misclassified examples.
RISK-02: Limited availability of free GPU resources	Free GPU services may have usage limits or availability issues, delaying BERTurk fine-tuning and model evaluation.	Start with the TF-IDF + Logistic Regression baseline, monitor available GPU resources, and use smaller batch sizes or reduced training configurations if necessary. Verify that the baseline remains functional independently of GPU availability.
RISK-03: Class imbalance	The dataset may contain significantly more 4- and 5-star reviews than 1-, 2-, and 3-star reviews. This may cause the model to favor the majority class (Positive) and perform poorly on minority classes, negatively affecting macro-F1.	Inspect the class distribution after mapping star ratings to sentiment labels. Apply class weights during model training or consider undersampling/oversampling if necessary. Evaluate per-class precision, recall, and F1-score, as well as macro-F1, using the original held-out test set.

19. Data and Storage Design

Input Data

The system accepts Turkish e-commerce customer reviews as text input. The dataset contains review texts and associated star ratings, which will be mapped to three sentiment classes during preprocessing.

Data Storage

The application uses SQLite to store sentiment analysis results locally. The database contains one main table named `analysis_results`.

Field	Data type	Description
id	INTEGER	Unique identifier for each analysis record (Primary Key, Auto Increment).
review_text	TEXT	Original Turkish customer review submitted by the user.
predicted_class	TEXT	Predicted sentiment class: Positive, Neutral, or Negative.
timestamp	DATETIME	Date and time when the analysis was performed.

External Dependencies

- Hugging Face: used to download the pre-trained BERTurk model and tokenizer.
- Python libraries: used for data processing, model training, and API development.
- SQLite: used for local data storage.

Privacy and Security Considerations

Customer reviews may contain personal or identifying information. The prototype runs locally and does not require user accounts or cloud-based storage. Only the information necessary for sentiment analysis and result storage will be retained. Sensitive personal information will not be intentionally collected, and the dataset usage conditions will be checked before use.

20. References

ACADEMIC REFERENCES

1. Altinok, D. (2025). Introducing TrGLUE and SentiTurca: A Comprehensive Benchmark for Turkish General Language Understanding and Sentiment Analysis. arXiv:2512.22100. https://arxiv.org/abs/2512.22100

   This paper introduces benchmarks for evaluating Turkish language models on natural language understanding and sentiment analysis tasks. I learned that these benchmarks can help researchers compare different models using standardized evaluation methods. I can use this study to guide my model evaluation and investigate whether its published results are comparable to my own experiments.

2. İncidelen, M., & Aydoğan, M. (2025). Sentiment Analysis in Turkish Using Language Models: A Comparative Study. European Journal of Technique, 15(1), 68–74. https://doi.org/10.36222/ejt.1592448

   This study compares several language models fine-tuned on 150,000 Turkish e-commerce reviews from the TRSAv1 dataset. It reports performance metrics such as accuracy and F1-score, providing a useful reference for selecting a Turkish language model and comparing it with my TF-IDF and Logistic Regression baseline.

3. Arzu, M., & Aydoğan, M. (2023). Comparative Analysis of Transformers Based Architectures for Turkish Sentiment Classification. Journal of Computer Science, IDAP-2023, 1-6. https://doi.org/10.53070/bbd.1350405

   I learned that Transformer-based models can effectively classify Turkish customer reviews into positive, neutral, and negative categories. I will use this study to understand how BERTurk-based architectures perform in Turkish sentiment classification and to support my model selection and evaluation process.

TECHNICAL REFERENCES

4. Hugging Face. Transformers Documentation. https://huggingface.co/docs/transformers (Accessed: October 2026)

   This documentation explains how to load pretrained language models, fine-tune them for text classification, and evaluate their predictions. I will use it to implement and evaluate the fine-tuned Turkish encoder in my customer review analysis service.

5. Ramírez, S. FastAPI Documentation. https://fastapi.tiangolo.com/ (Accessed: October 2026)

   This documentation explains how to build APIs, validate input data, and test application endpoints. I will use it to develop the API that receives customer reviews, sends them to the sentiment analysis model, and returns the prediction to the user.

6. scikit-learn. scikit-learn: Machine Learning in Python. https://scikit-learn.org/stable/ (Accessed: November 2026)

   I learned that scikit-learn provides tools for text feature extraction, machine learning model training, and model evaluation. I will use this library to build my TF-IDF and Logistic Regression baseline model and evaluate its performance using metrics such as accuracy and F1-score.